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- **PYTORCH**

PyTorch is an open-source machine learning and deep learning platform that was created by Meta company. PyTorch is a very convenient platform for developing, training, and deployment of deep learning models. PyTorch has automatic differentiation for backpropagation, optimal GPU utilization, and scalability for CPU and GPU architectures. In this project, PyTorch is used in combination with YOLOv11s to detect defects on the PCB, perform component identification, and perform inference operations in order to improve the inspection performance.

- **ULTRALYTICS YOLOv11**

YOLO v11 by Ultralytics is a popular computer vision object detection framework that efficiently detects objects as well as their classes. In the current project, there are two different models applied, namely YOLOv11s for PCB defect detection and YOLO11n for electronic component detection. These models detect bounding boxes, classes, and confidence score of the detected objects. The PCB defect detection model can detect six different classes of defects in PCBs, namely open circuits, short circuits, mousebites, spurs, spurious copper, and pin-holes.

- **OPENCV**

The Open Source Computer Vision Library, popularly referred to as OpenCV, is an open-source image and computer vision library that is extensively used in real-time applications. In this particular project, OpenCV is used in the process of PCB image pre-processing and transformation. OpenCV provides various functionalities to enhance, resize, and handle images in the pre-processing stage of the PCB image. The OpenCV library is further used in contrast limited adaptive histogram equalization (CLAHE) and high resolution tiling for small defect detection. The library makes it possible to integrate the code with both Python and PyTorch, thereby supporting the SmartPCB-XAI inspection process.